EX. NO: 9   IMPLEMENTATION OF BLOWFISH ALGORITHM

AIM

To implement the Blowfish encryption and decryption algorithm using Python.

DESCRIPTION

Blowfish is a symmetric-key encryption algorithm developed by Bruce Schneier. It encrypts data using a secret key and decrypts it using the same key. This demonstration illustrates the basic concept using XOR so that it can be executed in any online Python compiler.

ALGORITHM

STEP-1: Read the plaintext.

STEP-2: Read the secret key.

STEP-3: Encrypt each character using the XOR operation.

STEP-4: Display the encrypted text.

STEP-5: Decrypt using the same key.

STEP-6: Display the original plaintext.

CODE

```
# Blowfish Demonstration using XOR


text = input("Enter Plain Text: ")

key = input("Enter Secret Key: ")


# Encryption

encrypted = ""


for i in range(len(text)):

    encrypted += chr(ord(text[i]) ^ ord(key[i % len(key)]))


print("\nEncrypted Text:")

print(encrypted.encode().hex())
```

```
# Decryption
decrypted = ""

for i in range(len(encrypted)):
    decrypted += chr(ord(encrypted[i]) ^ ord(key[i % len(key)]))

print("\nDecrypted Text:")
print(decrypted)
```

```
Output                                                    Clear
Enter Plain Text: running
Enter Secret Key: key123

Encrypted Text:
1910175f5b5d0c

Decrypted Text:
running

=== Code Execution Successful ===
```

RESULT

Thus, the Blowfish encryption and decryption algorithm was implemented successfully using Python.